IMPERIAL COLLEGE LONDON

EE4-10
EE9-CS5-1
EE9-SC3
EE9-FPN2-02

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

PROBABILITY AND STOCHASTIC PROCESSES

Monday, 18 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Angeli

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Information for students

Each of the four questions has 25 marks.

The Questions

1. Random variables.

   a) Consider an experiment of flipping a fair coin.

      i) Let Y be a random variable such that $Y = 1$ if the outcome is "head" and $Y = 0$ otherwise. What is the expected value of Y? [6]

      ii) What is the probability of having j heads knowing that you have flipped the coin for n times? [6]

      iii) Suppose that we flip a coin until the first heads comes up. What is the probability distribution of the number of flips? [6]

   b) Suppose we wish to generate samples of a random variable with cumulative distribution function $F(x)$. This can be done by using an inverse transform: firstly generate random variable U uniform in the interval [0, 1], then apply inverse transform to obtain $X = F^{-1}(U)$ where F^{-1} denotes the inverse of F. Show that X has the correct distribution.

   [7]

2. Estimation and sequences of random variables.

a) The random variable X has the density $f(x) \sim c^4 x^3 e^{-cx}, \ x > 0$. We observe the i.i.d. samples x_i = 3.1, 3.7, 4.3, 4.9. Find the maximum-likelihood estimate of parameter c.

[10]

b) Let X be a Gaussian random variable with zero mean and variance σ^2. Estimate the tail probability $P(|X| > a)$ where $a = 5\sigma$ using

i) Markov inequality [5]

ii) Chebyshev inequality [5]

iii) Chernoff bound [5]

Hint: $E(|X|^n) = \begin{cases} 1 \cdot 3 \cdots (n-1)\sigma^n, & n \text{ even}, \\ 2^k k! \sigma^{2k+1} \sqrt{2/\pi}, & n = (2k+1), \text{ odd}. \end{cases}$

3. Random processes.

a) The number of failures $N(t)$, which occur in a computer network over the time interval $[0, t)$, can be modelled by a Poisson process $\{N(t), t \geqslant 0\}$. On average, there is a failure after every 4 hours, i.e. the intensity of the process is equal to $\lambda = 0.25$.

i) What is the probability of at most 1 failure in [0, 4), while at the same time at least 2 failures occur in [4, 8), and at most 1 failure in [8, 12) ? (time unit: hour) [10]

ii) What is the probability that the third failure occurs after 4 hours? [5]

Formula: Poisson distribution $P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k = 0,1,2,\cdots,\infty$

b) Consider the matched filter shown in the following figure:

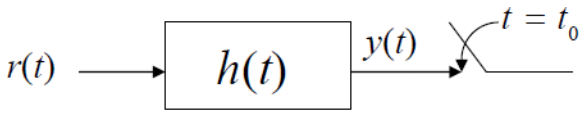


Fig. 3.1. Matched filter.

Let $r(t)$ represent a deterministic signal $s(t)$ corrupted by noise

$$r(t) = s(t) + w(t), \qquad 0 < t < t_0 = 1$$

where $w(t)$ is additive white noise with power spectral density $N_0 = 1$. Let $s(t)$ be a sinusoidal signal

$$s(t) = \begin{cases} \cos(2\pi t), & 0 < t < 1 \\ 0, & \text{otherwise} \end{cases}$$

i) Find the expression of the optimum matched filter $h(t)$. [5]

ii) Find the maximum output signal-to-noise ratio (SNR). [5]

4. Markov chains and martingales.

a) Let p be the probability that the coin turns "head", and let $q = 1 - p$. Gambler's ruin probability is given by

$$P_i = \begin{cases} \dfrac{1-\left(p/q\right)^{N-i}}{1-\left(p/q\right)^{N}}, & \text{if } p \neq \dfrac{1}{2} \\ \dfrac{N-i}{N}, & \text{if } p = \dfrac{1}{2} \end{cases}$$

A coin is flipped by a gambler with initial money £10. He wins £1 if the coin turns "head", and loses £1 if the coin turns "tail". The gambler will stop playing when he either has lost all of his money or he reaches £50. Of the first 10 flips, there are 6 "heads" and 4 "tails".

Calculate the probability that he will lose all of his money, given the results of the first 10 flips,

i) if the coin is fair, i.e., $p = 1/2$; [5]

ii) if the coin is unfair with $p = 1/3$. [5]

b) Fix a parameter $\lambda \in (0, 1)$ and let $X_0, X_1, X_2, \ldots$ be a sequence of independent random variables, whose distribution satisfies $P(X_j = -1) = P(X_j = 1) = 1/2$. Consider the following random sequence

$$Y_n = \sum_{i=0}^{n} X_i \lambda^i, \qquad n = 0,1,2,3,\ldots$$

i) Show $\{Y_n\}$ is a martingale. [3]

ii) Derive the characteristic function of Y_n. [4]

iii) Show that for any set E

$$P(Y_{n+1} \in E) = \frac{1}{2} P(Y_n \in T_1^{-1}(E)) + \frac{1}{2} P(Y_n \in T_2^{-1}(E))$$

where $T_1(x) = \lambda x + 1$ and $T_2(x) = \lambda x - 1$. [4]

iv) What is the limiting distribution of Y_n as $n \to \infty$ if $\lambda = 1/2$? [4]